Maximum Crown Width Prediction Equation

Greg Johnson, Aaron Weiskittel, David Marshall

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Maximum Crown Width (`MCW`) Prediction

Data

We used the Forest Health Monitoring data supplied by Aaron. It contains 127710 observations on 221 species. Earlier analysis indicated that trees with crown ratios (`cr`) below 0.5 were suspect for determining maximum crown width. We filtered the data to exclude low crown ratios (`crs`), yielding 71255 observations on 215 species. Additionally, we computed the sample size by species and filtered species with less than 200 observations, yielding 64328 observations on 65 species.

Aaron and David previously determined that the 95th percentile of crown width (`cw`) was a useful approximation for maximum crown width (`mcw`).

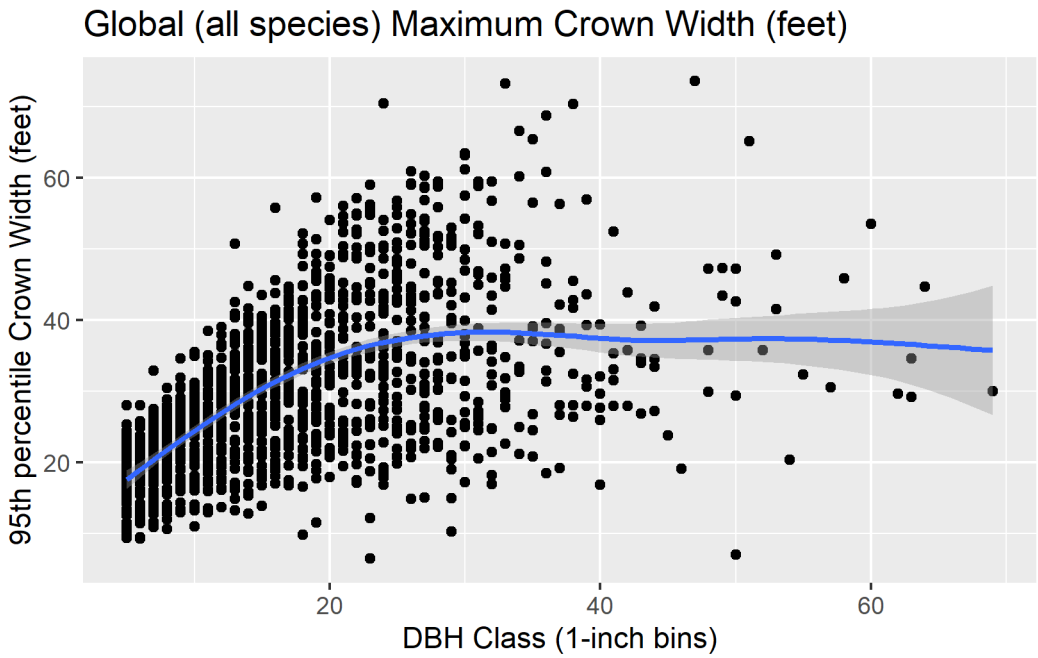
We split the analysis in two phases:

1. Develop equation form (system) and initial parameter estimates by computing the 95th percentile `cw` for 1-inch diameter at breast height (`dbh`) classes for each species. We constrained the data set to include `dbh` classes with 10 or more observations.
2. Fit the final model with quantile regression on the filtered data.

Phase 1

Global Classed `mcw`

Below is a graph of `mcw` (approximated using species-specific 1-inch `dbh` classes' 95th percentile `cw`). The graph illustrates that over all species, crown width tends toward an asymptote.



We fit the filtered and classed data using Equation 1:

$$cw_{95} = \beta_0 dbhc^{\beta_1} \tag{1}$$

where:

- cw_{95} - 95th percentile crown width (feet) for species-specific 1-inch `dbh` class
- $dbhc$ - 1-inch `dbh` class (inches)

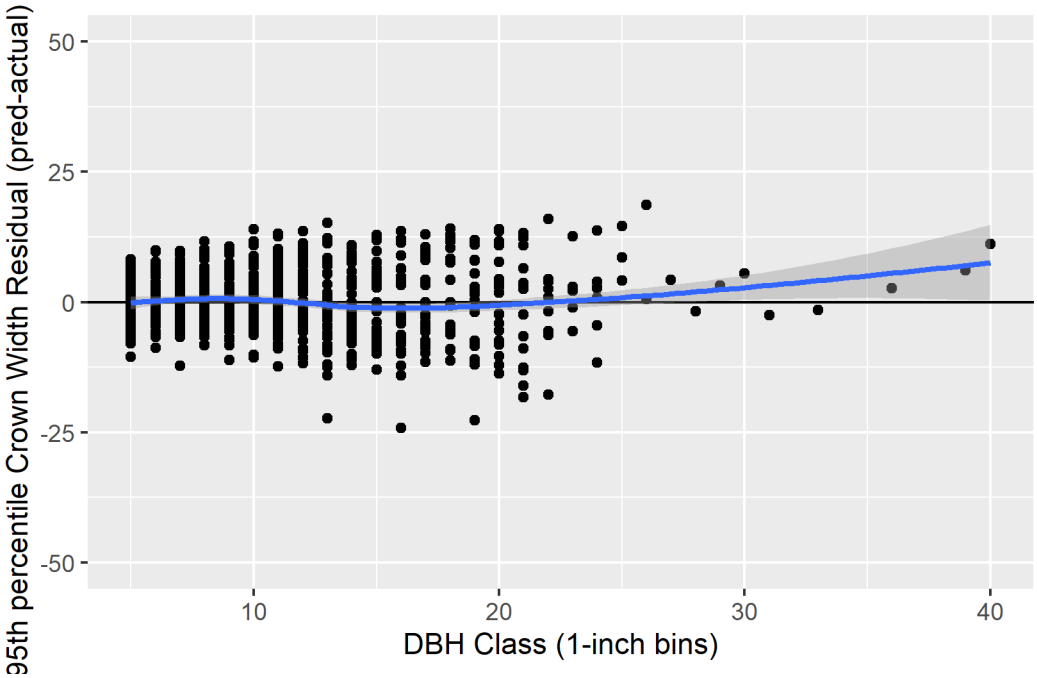
The fit statistics are shown below with a residual graph. The global fit does not show serious residual trends across `dbh`, but there is a significant difference between conifers and hardwoods.

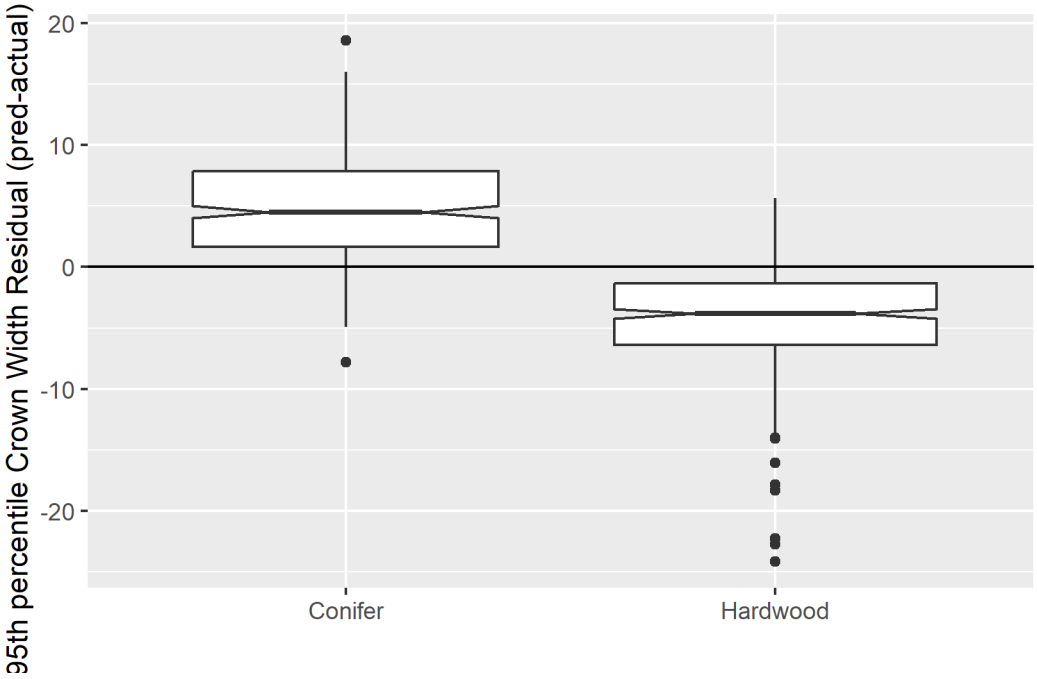
```
Formula: cw_95 ~ b0 * dbh_class^b1

Parameters:
   Estimate Std. Error t value Pr(>|t|)
b0  7.67714    0.37917   20.25   <2e-16 ***
b1  0.51022    0.01903   26.81   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 6.293 on 803 degrees of freedom

Number of iterations to convergence: 4
Achieved convergence tolerance: 3.204e-06
```





Global Classed `mcw` with Conifer/Hardwood Differentiation

We adjusted Equation 1 to add an indicator variable (`type` (1 if conifer, 0 if hardwood) to adjust β_0 and β_1. We found that the β_0 adjustment was not significant, so the resulting new equation is Equation 2:

$$cw_{95} = \beta_0 dbhc^{(\beta_1 + \beta_{1a} type)} \tag{2}$$

where:

- cw_{95} - 95th percentile crown width (feet) for species-specific 1-inch `dbh` class
- $dbhc$ - 1-inch `dbh` class (inches)
- $type$ - 1 if conifer, 0 if hardwood

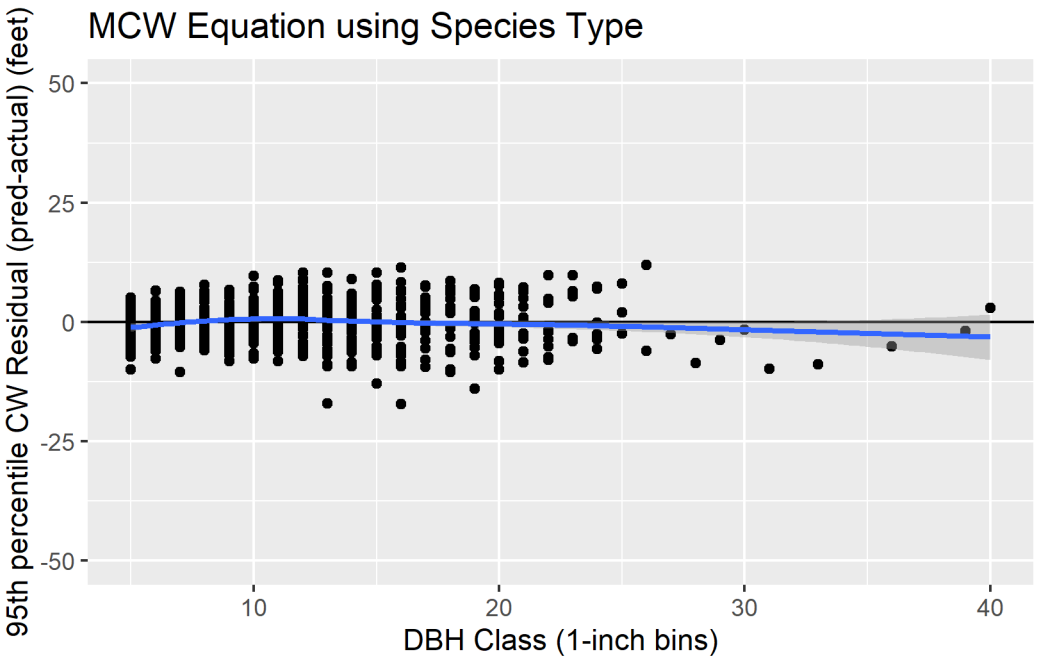
The fit statistics for Equation 2 are shown below with a residual graph. The new global fit does not show serious residual trends across `dbh`, and removes the difference between conifers and hardwoods. Looking at residuals at the species level reveals that there are species-level effects that need to be accounted for (see last graph in this section).

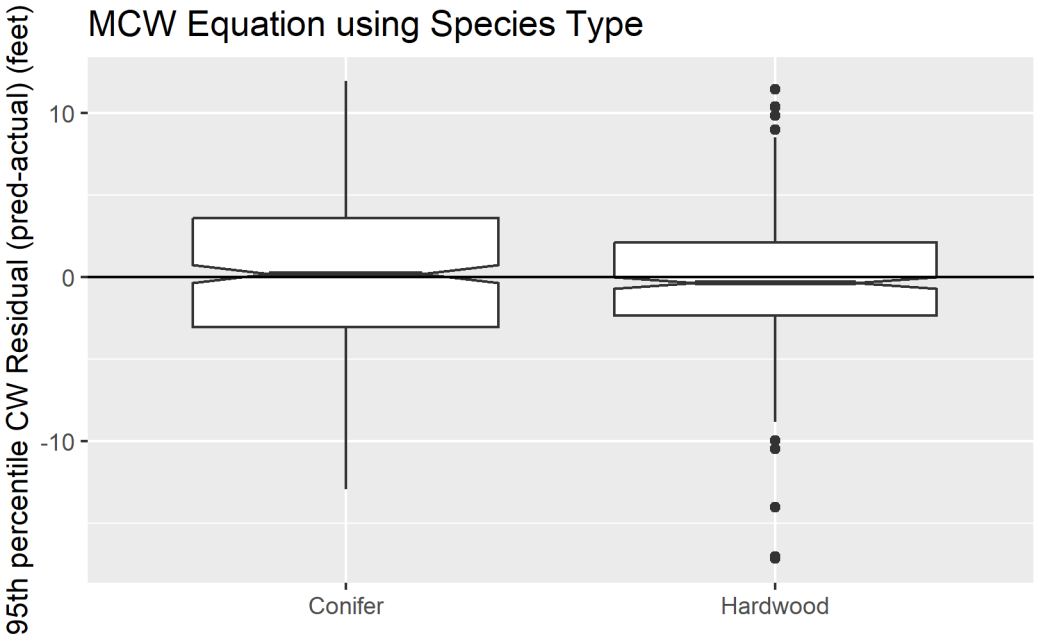
```
Formula: cw_95 ~ (b0) * dbh_class^(b1 + b1a * (type == "Conifer"))

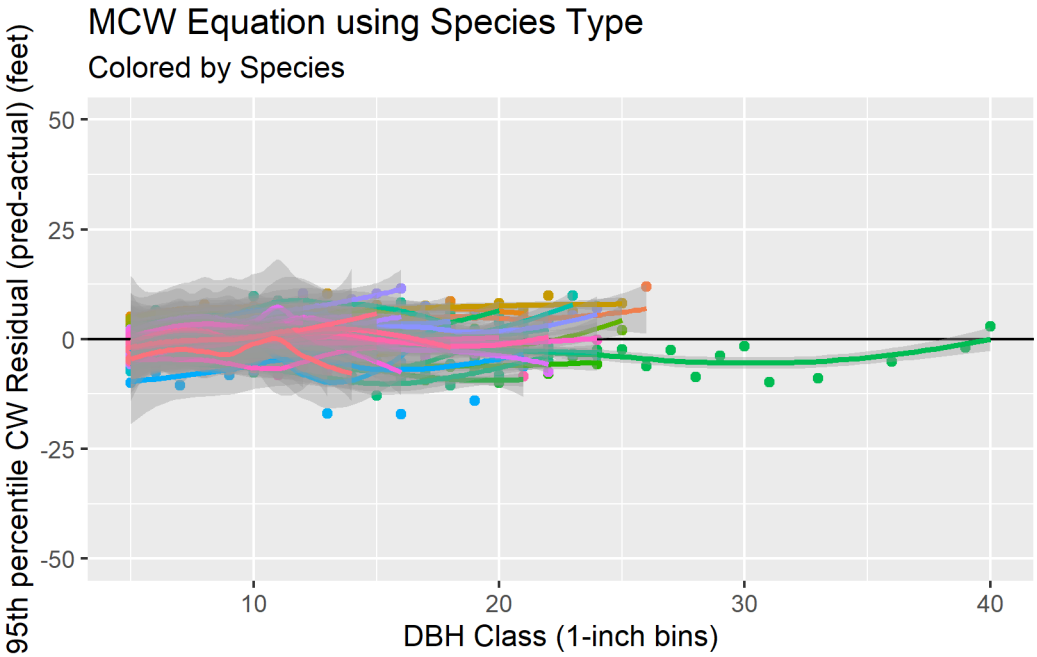
Parameters:
    Estimate Std. Error t value Pr(>|t|)
b0   6.28528    0.21372   29.41   <2e-16 ***
b1   0.65462    0.01346   48.63   <2e-16 ***
b1a -0.13807    0.00431  -32.04   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.119 on 802 degrees of freedom

Number of iterations to convergence: 4
Achieved convergence tolerance: 1.49e-08
```


MCW Equation using Species Type
95th percentile CW Residual (pred-actual) (feet)
50
25
0
-25
-50
10
20
30
40
DBH Class (1-inch bins)


MCW Equation using Species Type
95th percentile CW Residual (pred-actual) (feet)
10
0
-10
Conifer
Hardwood


MCW Equation using Species Type
Colored by Species
95th percentile CW Residual (pred-actual) (feet)
50
25
0
-25
-50
10
20
30
40
DBH Class (1-inch bins)

Phase 2

Parameter Estimates using Quantile Regression

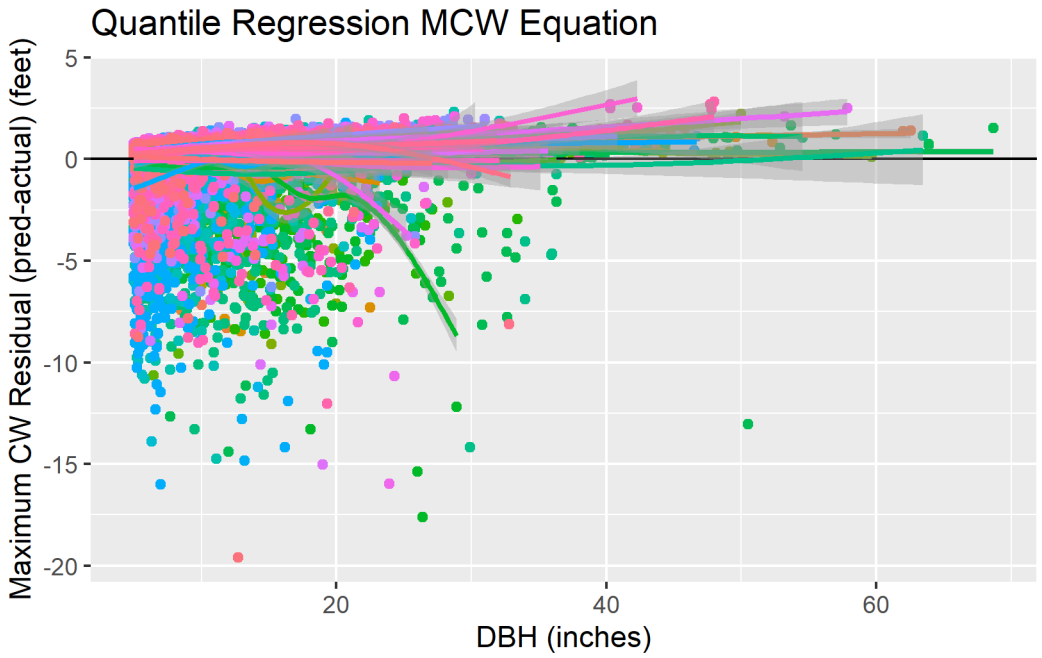
Using the filtered `cw` data we fit Equation 2 to the individual tree data using quantile regression with a $\tau = 0.95$. The predictions from the quantile regression.

The regression summary statistics for the quantile regression are shown below.

```
Call: quantreg::nlrq(formula = cw_ft ~ b0 * dbh_in^(b1 + b1a * (type ==
    "Conifer")), data = crown_width_data2, start = c(b0 = 5.3247,
    b1 = 0.435, b1a = -0.1362), tau = 0.95, trace = FALSE, control = list(
    maxiter = 100, k = 2, InitialStepSize = 1, big = 1e+20, eps = 1e-07,
    beta = 0.97))

tau: [1] 0.95

Coefficients:
    Value      Std. Error t value    Pr(>|t|)
b0    6.91138   0.07456   92.69779   0.00000
b1    0.64847   0.00498  130.08869   0.00000
b1a  -0.13670   0.00189  -72.44703   0.00000
```



species-level Parameter Estimates using Quantile Regression Estimates

We took the quantile regression parameter estimates above and fit a β_0 estimate by species also using quantile regression with a $\tau = 0.95$.

The residual graph using the species adjustments appears below, followed by a table of final parameter estimates.

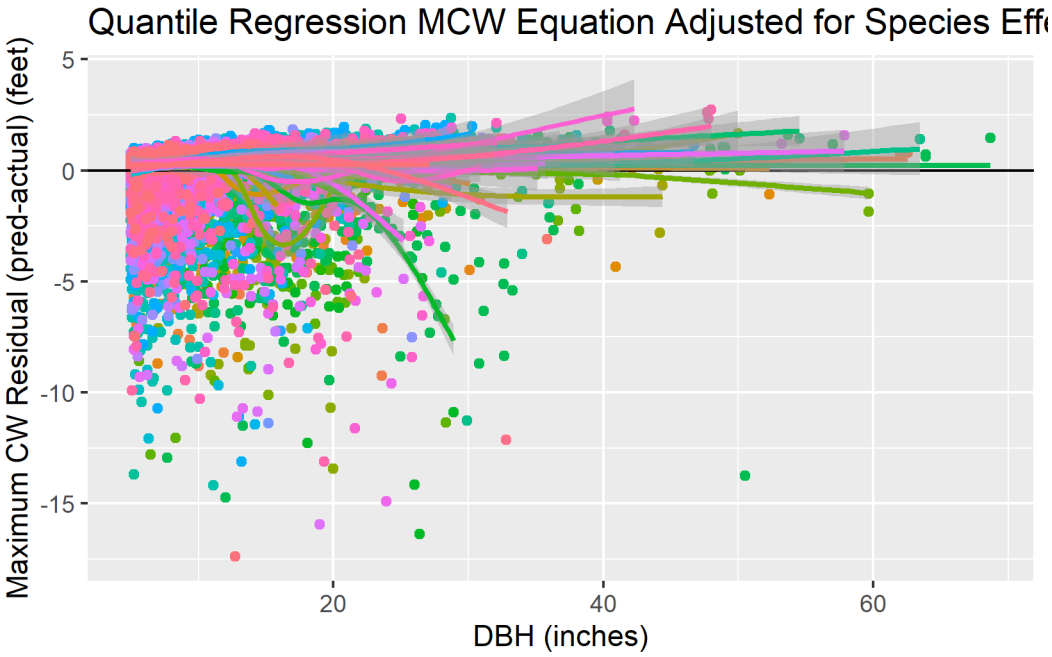


Table 1: Quantile Regression Estimates by Species

spcd	b0	b1	b1a
11	5.875762	0.6484729	-0.1366974
12	6.378250	0.6484729	-0.1366974
15	5.460375	0.6484729	-0.1366974
17	6.937084	0.6484729	-0.1366974
19	4.747365	0.6484729	-0.1366974
20	4.568566	0.6484729	-0.1366974
68	7.701941	0.6484729	-0.1366974
71	6.929172	0.6484729	-0.1366974
81	5.436014	0.6484729	-0.1366974
93	4.876252	0.6484729	-0.1366974
94	6.695981	0.6484729	-0.1366974
95	5.548966	0.6484729	-0.1366974
97	6.831443	0.6484729	-0.1366974

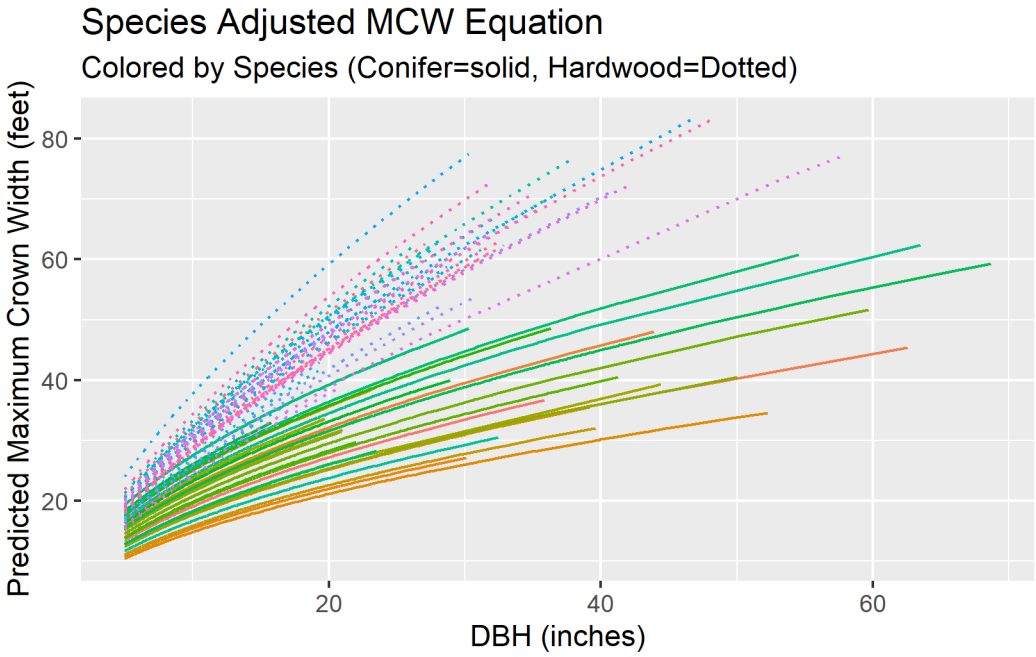
spcd	b0	b1	b1a
101	5.636916	0.6484729	-0.1366974
105	7.299473	0.6484729	-0.1366974
108	5.473901	0.6484729	-0.1366974
111	6.648275	0.6484729	-0.1366974
116	6.373056	0.6484729	-0.1366974
122	6.034930	0.6484729	-0.1366974
125	6.107467	0.6484729	-0.1366974
129	7.728772	0.6484729	-0.1366974
131	7.154805	0.6484729	-0.1366974
132	8.020682	0.6484729	-0.1366974
202	6.812357	0.6484729	-0.1366974
241	5.618032	0.6484729	-0.1366974
242	7.857309	0.6484729	-0.1366974
261	8.469442	0.6484729	-0.1366974
263	7.450865	0.6484729	-0.1366974
264	5.139181	0.6484729	-0.1366974
313	6.837342	0.6484729	-0.1366974
316	7.035818	0.6484729	-0.1366974
318	7.257058	0.6484729	-0.1366974
351	7.227343	0.6484729	-0.1366974
371	7.168883	0.6484729	-0.1366974
372	7.489065	0.6484729	-0.1366974
375	6.113663	0.6484729	-0.1366974
400	6.616473	0.6484729	-0.1366974
403	6.871574	0.6484729	-0.1366974
407	6.663411	0.6484729	-0.1366974
409	7.252353	0.6484729	-0.1366974
531	8.482795	0.6484729	-0.1366974
541	6.876362	0.6484729	-0.1366974
543	5.744789	0.6484729	-0.1366974
591	6.294185	0.6484729	-0.1366974
611	5.996279	0.6484729	-0.1366974
621	6.435790	0.6484729	-0.1366974
631	5.833417	0.6484729	-0.1366974
693	7.078069	0.6484729	-0.1366974
711	6.903003	0.6484729	-0.1366974
746	5.415871	0.6484729	-0.1366974
762	6.828212	0.6484729	-0.1366974
802	6.766737	0.6484729	-0.1366974
805	5.559936	0.6484729	-0.1366974
806	7.056191	0.6484729	-0.1366974

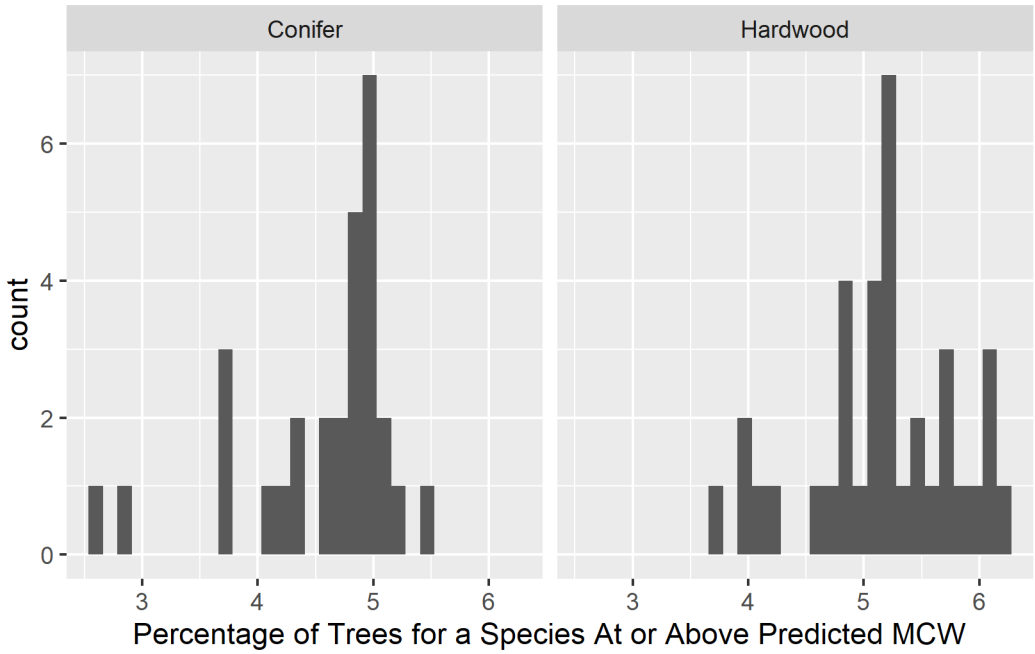
spcd	b0	b1	b1a
807	6.470068	0.6484729	-0.1366974
812	7.064190	0.6484729	-0.1366974
818	6.396066	0.6484729	-0.1366974
823	6.368065	0.6484729	-0.1366974
827	7.709675	0.6484729	-0.1366974
832	6.587478	0.6484729	-0.1366974
833	6.745180	0.6484729	-0.1366974
835	6.429724	0.6484729	-0.1366974
837	6.481113	0.6484729	-0.1366974
951	6.471522	0.6484729	-0.1366974
972	7.359866	0.6484729	-0.1366974

Equation Performance

Below is a graph of equation predictions by species, followed by a histogram of the percentage of all trees with `cr` > 0.5 that have `cw_ft` $\geq \hat{mcw}$. We would expect the percentages to cluster near 5% given our τ of 0.95.

Five species fall far outside our expectations and have reasonably large sample sizes.





Conclusion

The system developed above accounts for species-level variation in `cw` without sacrificing biologically meaningful equation performance. It is unbiased against the fitting data. Generally, hardwoods have wider crowns than conifers.

The system does produce some abhorrent predicted crown widths that should be investigated.